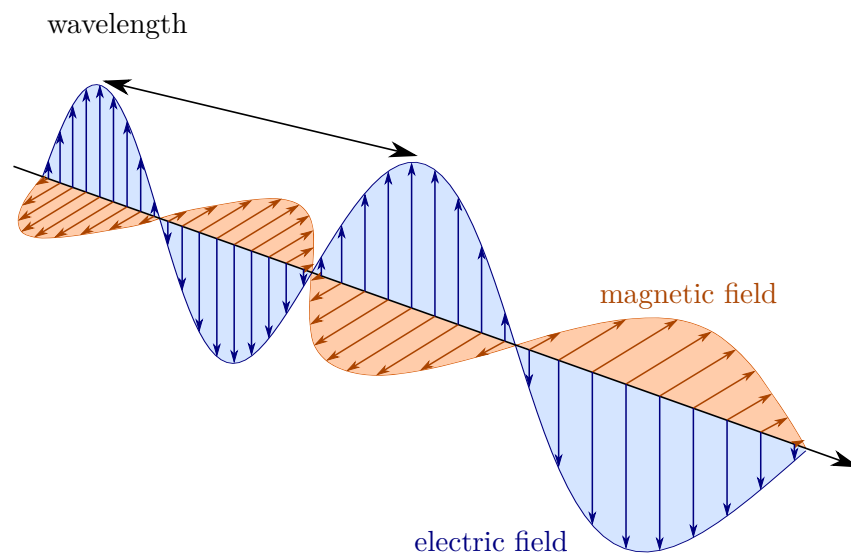

Electromagnetic Theory

TAMU ECEN-635

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Acknowledgements

These notes were compiled from the course materials for ECEN-635 (Electromagnetic Theory) taught by Dr. Robert Nevels at Texas A&M University during the Fall 2025 semester.

Electromagnetic Theory (ECEN-635)

 **Instructor:** Dr. Robert Nevels
 **Semester:** Fall 2025
 **Time:** ONLINE
 **Location:** ONLINE

Description: Maxwells equations, boundary conditions, Poyntings theorem, electromagnetic potentials, Greens functions, Helmholtzs equation, field equivalence theorems; applications to problems involving transmission scattering and diffraction of electromagnetic waves.

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Chapter 1

Introduction

The purpose of this course is to learn some of the basic properties of high frequency electromagnetic fields and circuit analysis techniques. Some major areas and applications of electromagnetics include:

1. Scattering and propagation
 - Wireless device coverage area
 - Radar
 - EMI (interference), EMC (compatibility)
2. Antennas
 - Wires and wireless devices
 - Autos, aircraft
 - Magnetic resonance
3. High frequency circuits
 - Microwave analog circuits (5G, 6G)
 - Compute high pulse rate digital circuits
 - Nanotechnology
4. Optics
 - Circuits
 - Reflection and refraction

Maxwell's equations are the foundation of electromagnetics. They are fundamental to our world, and are defined in time domain point form as

$$\nabla \times \mathcal{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathcal{B}(\mathbf{r}, t) - \mathcal{M}(\mathbf{r}, t) \quad (1.1)$$

$$\nabla \times \mathcal{H}(\mathbf{r}, t) = \frac{\partial}{\partial t} \mathcal{D}(\mathbf{r}, t) + \mathcal{J}(\mathbf{r}, t) \quad (1.2)$$

$$\nabla \cdot \mathcal{D}(\mathbf{r}, t) = \rho(\mathbf{r}, t) \quad (1.3)$$

$$\nabla \cdot \mathcal{B}(\mathbf{r}, t) = \rho_m(\mathbf{r}, t) \quad (1.4)$$

and then there was light.

By taking the divergence of (1.2), we see

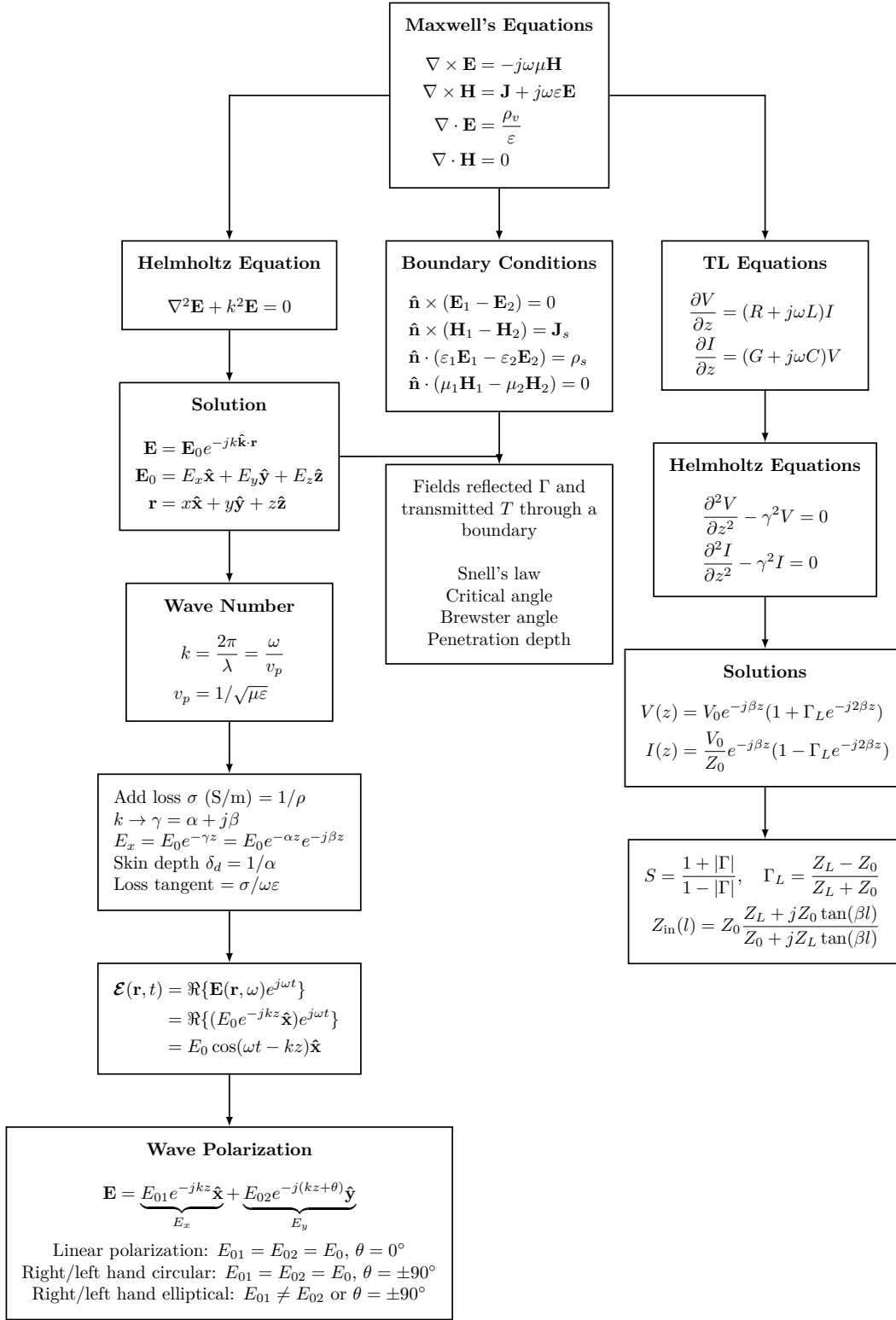
$$\begin{aligned} \nabla \cdot [\nabla \times \mathcal{H}(\mathbf{r}, t)] &= \frac{\partial}{\partial t} \nabla \cdot \mathcal{D}(\mathbf{r}, t) + \nabla \cdot \mathcal{J}(\mathbf{r}, t) \\ 0 &= \frac{\partial}{\partial t} \rho(\mathbf{r}, t) + \nabla \cdot \mathcal{J}(\mathbf{r}, t) \\ \therefore \quad \nabla \cdot \mathcal{J}(\mathbf{r}, t) &= -\frac{\partial}{\partial t} \rho(\mathbf{r}, t) \end{aligned} \quad (1.5)$$

This is known as the *equation of continuity*, which is a conservation law of electromagnetics. All accepted theories of physics are expected to have a continuity equation.

A **conservation between the time and frequency** (time harmonic) domains, would typically be written:

$$\mathcal{E}(t) = \Re\{\mathbf{E}e^{j\omega t}\} = \Re\{|\mathbf{E}|e^{j\alpha}e^{j\omega t}\} = |\mathbf{E}| \cos(\omega t + \alpha)$$

Converting Maxwell's equations to time harmonic form, we have

**Figure 1.1:** Flowchart of key equations and concepts in electromagnetic theory.

Electromagnetics flowchart

See the flowchart on the next page for a summary of key equations and concepts in electromagnetic theory.

